RAGHAVAN MADABUSHI

Atlanta, GA | 205-427-2717 | raghavanmadabushi10@gmail.com | U.S. Permanent Resident
linkedin.com/in/raghavanmadabushi | github.com/RaghavanMadabushi14 | Open to Relocation

EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Electrical Engineering *Aug 2025 – Dec 2028*
GPA: 3.80 / 4.00
Relevant Coursework: Programming Hardware/Software Systems in C, Computer Architecture, Digital System Design, Circuit Analysis, Signal Processing, Object-Oriented Programming, Differential Equations, Multivariable Calculus

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog, Bash, MATLAB, Java
Embedded & Firmware: Microcontrollers (Arduino, Teensy), Serial Communication Protocols, Motor Controllers, Firmware Validation, Hardware-in-the-Loop (HIL) Testing, USB Serial Telemetry, Real-Time Telemetry Pipelines
Hardware & Digital Design: RTL Design, FPGA Development (Xilinx Vivado), Computer Architecture, PCB Design (Altium), Power Management, Circuit Analysis
Robotics & Tools: ROS2 (Jazzy), Gazebo, MuJoCo, URDF, Linux, Git, GitHub Actions, CI/CD, Docker, pytest

EXPERIENCE

RealBench | getrealbench.dev Atlanta, GA
Founder & Sole Engineer – Georgia Tech CREATE-X Startup Accelerator *Jun 2026 – Present*
Startup building automated firmware validation and reset infrastructure for robotics test pipelines.

- Built a fail-closed readiness system in Python that automates the reset-and-verify cycle between robot firmware test runs, evaluating seven tiered checks against live ROS2, Gazebo, or snapshot telemetry and returning a three-state verdict (ready / recoverable / quarantine) with distinct exit codes for CI pipeline gating.
- Designed the core safety invariant so unverifiable hardware state can never certify a robot as ready, then proved it by exhaustively enumerating every check-state combination across a 128-test, negative-path-dominant suite.
- Engineered a six-case ROS2 integration harness against a live Jazzy node in Docker, surfacing and fixing two latent defects, including a fail-open path that counted joints reporting null positions as present.
- Validated the system against six production robot descriptions (Unitree A1, Unitree G1 humanoid, ANYmal B, Franka Panda, Champ, TurtleBot3), scaling from 12 to 29 actuated joints with zero model-specific configuration.
- Interviewed 50+ robotics engineers across field, aerial, legged, and exoskeleton platforms, quantifying 5–30 min of reset downtime per cycle.

NurturIQ Atlanta, GA
Founder & CTO – Georgia Tech CREATE-X Startup Accelerator (pivoted to RealBench) *Dec 2025 – Jun 2026*

- Engineered a 120 FPS eye-tracking pipeline in ARKit to map gaze trajectories in real time on consumer mobile hardware.
- Shipped the core diagnostic engine, an ML recommender matching user profiles against a 1,700-exercise dataset.

University of Alabama at Birmingham – Cardiovascular Research Laboratory Birmingham, AL
Research Assistant *Summer 2023*

- Built and calibrated experimental instrumentation for cardiovascular research trials, executing measurement protocols and recording quantitative data across study sessions.

LEADERSHIP & ACTIVITIES

RoboNav (RoboJackets – Georgia Tech University Rover Challenge Team) Atlanta, GA
Electrical Subteam Member *Aug 2026 – Present*

- Programming Arduino and Teensy microcontrollers in C/C++ to drive DC motor controllers and implement serial communication protocols for the rover's electrical subsystem in the team's Mini Rover Challenge.
- Soldering, crimping, and assembling motor-driver and power-management circuitry; contributing to a custom PCB in Altium.

EnergyHack@GT, Lead Organizer Oct 2025 – Present
Coordinate a 100-team hardware and software energy hackathon backed by GE Vernova, ExxonMobil, and Kimley-Horn.
American Psychiatric Association Annual Conference, Selected Presenter May 2026
Presented technical research on real-time sensing and behavioral screening systems to a national professional audience.
Beta Theta Pi (Georgia Tech Chapter), Academics & Risk Management Chair Fall 2025 – Present
Own risk and safety protocols for a 100+ member chapter, maintaining zero safety infractions.